

SLRI-Cal [Embedded Mode] User Manual

The Persistent Reconstructi... — □ ×

SLRI-Cal [Embedded Mode]

Initialize Calibration

Select Measurement Mode

☒ Transmission Mode

☐ Fluorescence Mode

☐ Transmission and Fluorescence Mode

Select Calibration Mode

☒ Real-time Calibration to Latest Foil

☐ Relative Energy Alignment via in-line Reference

☐ Real-time Hybrid Alignment to First Sample

Element : Auto Select ▾

E0 [eV]: Auto Select

Foil Reference

Energy column : Auto Select ▾

$\mu(E)$ column : Auto Select ▾

Sample

Energy column : Auto Select ▾

Reference $\mu(E)$ column : Auto Select ▾

SLRI-Cal [Embedded Mode]

1. Software Overview & Objectives

SLRI-Cal [Embedded Mode] is an automated processing software designed to perform **real-time Energy Calibration** for XAS (X-ray Absorption Spectroscopy) experimental data files.

The system triggers automatically upon completion of a scan as soon as the spectral data file is generated from the beamline. It takes **only a few seconds per file** to perform energy alignment and drift calculations. Users only need to configure the Measurement Mode, Calibration Mode, and Watch Folder once during the initial setup. Following this, the system executes the entire workflow automatically.

- **Data Security & Integrity:** The system **never modifies or alters the original raw data files**. Instead, it generates a new calibrated file and saves it separately in a dedicated output folder.
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2. Step-by-Step Workflow

- ◆ Step 1: Select Measurement Mode

This step defines the reference column position of the data. Select the measurement mode that matches your current experimental setup:

- **Transmission Mode**
- **Fluorescence Mode**
- **Transmission and Fluorescence Mode**

- ◆ Step 2: Select Calibration Mode

This version offers **three calibration modes** based on the characteristics of your reference signals:

1 Real-time Calibration to Latest Foil

- **Working Principle:**
 1. Calibrate the latest incoming Foil.
 2. Shift new incoming files by the latest Foil offset.
- **Description:** The software calculates the energy drift from the latest detected Foil file. It then applies this energy offset to shift the energy axis of subsequent incoming Sample files into the correct position immediately.

- **Example:** If an initial Fe-Foil scan reveals an energy drift of +0.3 eV, subsequent Sample files will automatically be shifted by -0.3 eV. If a new Foil (e.g., Fe-foil or Cu-foil) is scanned midway through the experiment, the system updates and saves the new drift value immediately for the next batch of Sample files.

2 Relative Energy Alignment via In-line Reference

- **Working Principle:**
 1. Calibrate the latest incoming Foil.
 2. Align the In-line Reference of new files to the calibrated Foil.
- **Description:** This mode is ideal when the sample's In-line Reference signal has high resolution and sharp features. The system first evaluates the latest Foil file. It then directly aligns the In-line Reference signal of each new Sample file to the energy coordinates of the calibrated Foil on a file-by-file basis.

3 Real-time Hybrid Alignment to First Sample

- **Working Principle:**
 1. Calibrate the latest incoming Foil.
 2. Shift new Sample files by the latest Foil offset.
 3. Align the In-line Reference of new files to the 1st Sample's Reference.
- **Description:** This hybrid mode is best suited for scenarios where the sample's In-line Reference signal has low resolution. The system determines the drift from the latest Foil and applies that offset to shift the energy axis of the first new Sample file (the one chronologically closest to the foil). When subsequent Sample files (2nd, 3rd, 4th, etc.) arrive, the system automatically switches its logic: it aligns the In-line Reference signal of the latest Sample file to the In-line Reference of that very first Sample file.

◆ Step 3: Initialize Calibration

When you click the **Initialize Calibration** button, a window will prompt you to select the primary experimental data folder currently receiving live beamline measurements (**Watch Folder**).

Once confirmed, the system enters a continuous loop to monitor the folder. As soon as a measurement file is completed, it is processed and saved into a new dedicated directory named:

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The original raw files remain completely untouched.